NAME: ________________________ **PERIOD:** ________

One Interval, Two Threshold Claims

AP Statistics · Unit 3 · Topic 3.4

[STAR] THE PROBLEM

Suppose a statewide activities association has 10,000 member students and is considering requiring reusable food containers at its tournaments. In a simple random sample of 300 member students, 204 favor the requirement. A sponsor claims that at least 65% of all member students favor it; an event organizer makes the more modest claim that more than 60% favor it.

Kai: "The sample has $204/300 = 68\%$ in favor, above both thresholds, so it supports both claims."

Mira: "The point estimate alone cannot settle both claims. We should construct a confidence interval and compare all of its plausible values with each threshold."

Sample counts

Group	N	n	Favor	Other
Members	10,000	300	204	96

FIRST TAKE — one sentence before you work the parts

Do the data appear to support both claims, only one, or neither? Cite one number and commit before calculating the interval.

[BOOK] LET THE WHOLE INTERVAL SET THE BOUND (keep on the page)

For a one-sample z-interval for p, check random sampling, $n \leq 0.10N$, and at least 10 observed successes and failures. Then use $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ for 95% confidence. Interpret: "We are 95% confident that the interval captures the population proportion [in context]." A claim $p > c$ is supported only when the whole interval lies above c; if c is inside the interval, plausible values lie on both sides.

Work (a)–(c).

DOK 1 **(a)** Define the population proportion p and calculate the sample proportion $\hat{p}$.

DOK 2 **(b)** Verify the conditions, construct a 95 percent confidence interval for p, and interpret the interval in the association's context.

DOK 3 **(c)** In 3–4 sentences, decide which threshold claim the interval supports and whose reasoning is better. Compare the lower endpoint with 0.60 and 0.65. Explain why a plausible 0.65 is different from establishing at least 0.65.

Frame: The interval supports the claim ______________________ because its lower endpoint, _________, is __________________.

Frame: Although 0.65 is ________________, it does not establish ___________________________; Kai's argument would make readers think ____________________________.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.